CIS 635 Knowledge Discovery & Data Mining

Course Introduction: CIS 635 (KDD)

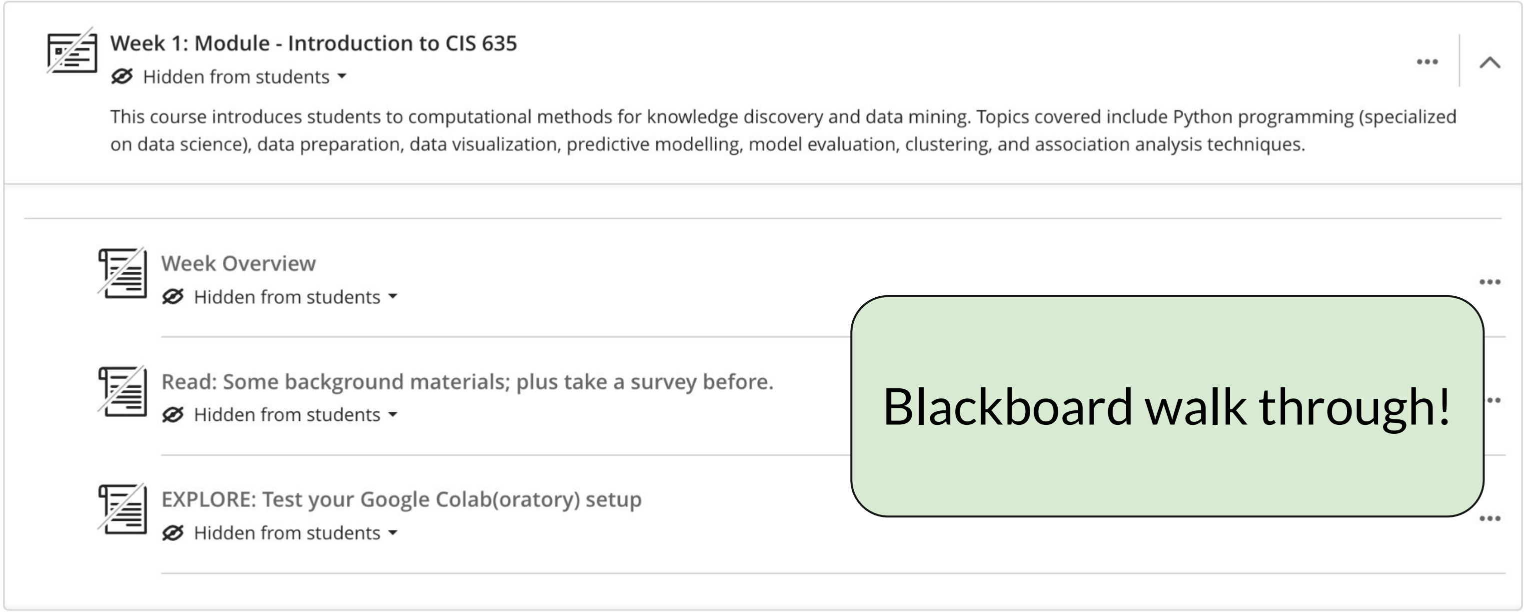
Communication Channels:

- ***Blackboard** messages (primary choice)*
- *Email*
- *In person/Online*

  By appointment (Booking Calendar):
  Mackinac Hall (MAK) D-2-216 (Allendale)

Pronouns: he/him/his
Email: hasanka@gvsu.edu;
hasanka@mail.gvsu.edu

Week 1: Introduction to CIS 635



Blackboard walk through!

Week overview

This week we will:

- Get to know each other (networking)
- Set up our course objective, guidelines, and evaluation procedure.
- Set up our programming development environment(s), more specifically, Google Colab(oratory) on your Google drive.
- Test source code and report delivery methods.

Due:

What you have to do is create and share a Jupyter Notebook on your Google Colab(oratory) environment.
Get a copy of this notebook and start playing with.

Networking!

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- **Probabilistic ML**
    - **Semi-supervised Learning**
    - **Generative AI** (example: LLMs ,ChatGPT)

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- **Last ~9 years as a Data Scientist; delivered several AI products in multiple domains**

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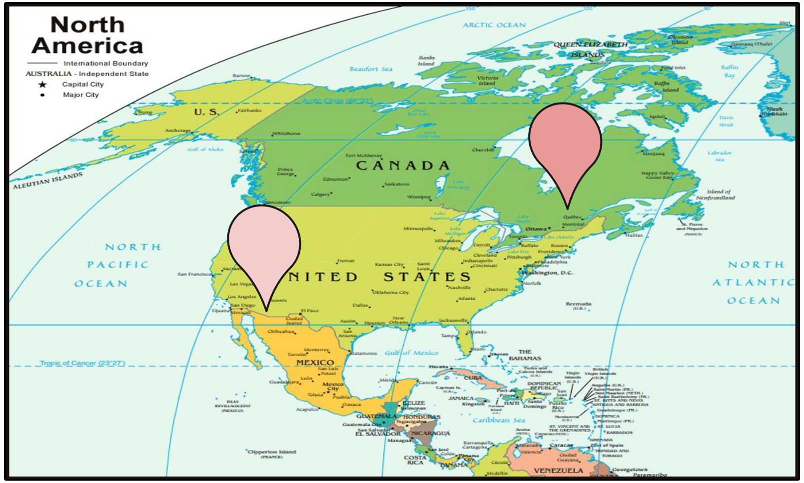

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- Travelling
- **Gardening**

About you!

- I hope I would get to know each of you as we progress
- We will meet in person and/or virtually as per our availability
- You will collaborate as groups
  - For discussion different topics: course specific and beyond
  - Final project
- I will seek your suggestions

Blackboard messages!

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General information (about the course)

Description: This course introduces students to

- Computational methods for knowledge discovery and data mining.
- Topics covered include
  - Python programming (specialized on data science),
  - EDA: data preparation, analysis, and visualization
  - Predictive modeling, model evaluation, clustering, time series data analysis, and special topics including deep learning

Objective: Completing this course, you should be able to:

- Wrangle, analyze, and visualize data.
- Exercise and apply supervised and unsupervised knowledge discovery techniques.
- Identify and build models using common data mining techniques.
- Evaluate models for their effectiveness and appropriateness.
- Communicate findings using effective visualizations.

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General information (about the course)

Textbook(s): There are no required textbooks for this course. Some good options include:

- **Data Mining** - A Knowledge Discovery Approach by Cis Pedrycz, and Swiniarski (available online through the GVSU library)
- Introduction to Data Mining Concepts and Techniques, 2nd Edition by Tan, Steinbach, and Kumar
- **Pattern Recognition** and Machine Learning by Christopher M. Bishop, 2016
- **Machine Learning** with **Python Cookbook**: Practical Solutions from Preprocessing to Deep Learning Paperback – April 17 2018 by Chris Albon
- Fundamentals of **Data Visualization** by Claus Wilke (available online)
- **Python for Data Analysis** by Wes McKinney (accessible online through library)
- Automate the Boring Stuff with Python: Practical Programming for Total Beginners Paperback, by AI Sweigart (accessible online through library)
- Python Data Visualization Essentials Guide: Become a Data Visualization expert by building strong proficiency in Pandas, Matplotlib, Seaborn, Plotly, Numpy, and Bokeh (English Edition) Paperback – July 30 2021, by Kalilur Rahman

Requirements: **Blackboard**, **Computer** access, ability to run **Python** (and install any requisite Python packages). If you have trouble, talk to the instructor, and we will work out a solution together.

Delivery Plan

General information:

- See **Blackboard** for detailed schedule
- Planning and delivery per week
- **Course materials** and **assignments** will be available on Blackboard **mostly** at the beginning of each week; sometimes as we progress.

Key dates:

- 8/31:: Class begins (weekly planning & delivery)
- 10/25-27: Spring Break
- 12/12: Class Ends
- 12/19: Semester ends

Update your Blackboard settings so you receive your emails, messages, and notifications in time.

Office: MAK D-2-216 (by appointment),
Meeting Times: Online or in person

Delivery Plan

Introduction (KDD & Python) and networking (2W)

Week	Week of	Topics
1	Aug. 31	Introduction to Knowledge Discovery & Data Mining; Python Basics
2	Sept. 7	Python for Data Science; Descriptive Statistics & Data Visualization
3	Sept. 14	Exploratory Data Analysis (EDA) I
4	Sept. 21	Exploratory Data Analysis (EDA) II & Case Studies
5	Sept. 28	Similarity & Distance Measures; K-Nearest Neighbors
6	Oct. 5	Predictive Modeling: Regression
7	Oct. 12	Predictive Modeling: Classification I
8	Oct. 19	Predictive Modeling: Classification II & Model Evaluation
9	Oct. 26	*Fall Break*
10	Nov. 2	Unsupervised Learning: Clustering
11	Nov. 9	Clustering & Dimensionality Reduction
12	Nov. 16	Sequence Analysis & Modeling
13	Nov. 23	Special Topics: Deep Learning & Neural Networks
14	Nov. 30	Special Topics: Emerging Data-Mining Techniques
15	Dec. 7	Final Project Presentations & Course Wrap-Up
16	Dec. 14	Final Examination & Evaluation Week

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Supervised Modeling (4W)

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Introduction (KDD & Python) and networking (2W)

EDA (2W)

Supervised Modeling (4W)

Unsupervised Learning (2W)

Sequence Analysis and Modeling (1W)

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Introduction (KDD & Python) and networking (2W)

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Supervised Modeling (4W)

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Sequence Analysis and Modeling (1W)

Special Topics: DL, NN

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Introduction (KDD & Python) and networking (2W)

EDA (2W)

Supervised Modeling (4W)

Unsupervised Learning (2W)

Sequence Analysis and Modeling (1W)

Special Topics: DL, NN

Project (Learning by Doing)

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Evaluation

Grading distribution:

- Exercises /assignments/tests: 30%
- Exam / quizzes: 40%
- Final project (group + individual performance): 30%

Grade points:

A	93%	**C**	73%
A-	90%	**C-**	70%
B+	87%	**D+**	67%
B	83%	**D**	60%
B-	80%	**F**	Below 60%
C+	77%		

Note: Your final grade percentage will be rounded to the next integer percentage value. For example, an **89.1%** will round up to a **90%.**

Midterm, Quizzes, and Final Exam Policy

Written

- *Midterm exam,*
- *two quizzes,*
- *Final Exam*

will assess the concepts in machine learning covered up to that point. Additional details regarding the format and content of these assessments will be provided as the course progresses. There will be a final test. This is mainly to check student's critical, mathematical thinking, and problem solving ability as per this course is taught.

To help maintain the integrity of the course, all exams—including quizzes—will be conducted using the **Respondus LockDown Browser** with the webcam enabled. We will share detailed instructions before each exam or quiz to guide you through the process. Please take a moment to ensure that the LockDown Browser is installed and set up ahead of time. You can find helpful instructions in the guide below, and feel free to reach out to me if you have any questions or need assistance—I'm happy to help.

https://services.gvsu.edu/TDClient/60/Portal/KB/ArticleDet?ID=9865

Policy & expectation

Expectation: I expect the following to ensure your success in this course:

- check Blackboard on a regular basis for announcements, course material, and assignments
- stay up to date with required course materials.
- let me know how the class and my teaching can be improved
- adhere to the **GVSU policy of Academic Honesty** http://www.gvsu.edu/coursepolicies/

Course policy:

- Weekly reflections and homework assignments are to be completed **individually**.
- Due dates: All assignments will be due at 11:59pm Michigan time on the due date (unless otherwise stated).
- Late policy: You will lose 10% off of your maximum grade per day late, to a cap of 3 days (30% off), after which the assignment will not be accepted.

Policy & expectation

Academic Honesty:

- All students are expected to adhere to the academic honesty standards set forth by GVSU.
- In addition, students are expected to adhere to the academic guidelines as set forth by the CoC: https://www.gvsu.edu/computing/academic-honesty-30.htm
- You can learn a lot from your peers, therefore, I encourage collaboration, but passing their work off as your own is prohibited.

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<u>With respect to all individual assignments in this course:</u>

- Document collaboration; **no electronic transfer of code** between students is permitted.
- You are encouraged to engage in conversations in **online forums**, but do not post solutions or solicit others to complete your work for you.
- You are encouraged to talk about problems with each other in **non-technical terms** (i.e., not code).
- Ultimately, **you are responsible** for all aspects of your submissions. You should be able to explain and defend if the work is entirely your own.

Policy & expectation

Use of Generative AI Tools for Learning

- Students are permitted to use generative AI tools (e.g., ChatGPT) to support their learning, such as clarifying concepts, generating test cases, or assisting with code debugging.
- However, AI must not be used to complete or generate any part of the final work submitted for credit.
- All AI use must be clearly documented, and students are responsible for verifying the accuracy of AI-generated content.
- Improper or undocumented use of AI may constitute an academic integrity violation.

For detailed usage guidelines, responsibilities, and documentation requirements, please refer to the full document: **"Guidelines for Using Generative AI Tools in Learning"**, shared through **Blackboard**.

Useful resources

Blackboard: Course materials, assignments, grades, and announcements will be posted to Blackboard (https://lms.gvsu.edu/). It is your responsibility to stay informed.

Other academic resources: GVSU also provides opportunities for students to improve your **academic skills** through resources, such as:

- The writing center
- Computing Success Center
- Speech lab
- Research consultants
- Library Resources

Disability support : If you are in need of accommodations due to disability you must present a memo to me from Disability Support Resources (DSR), indicating the existence of a disability and the suggested reasonable accommodations. If you have not already done so, please contact the Disability Support Resources office (215 CON) by calling 331-2490 or email to dsrgvsu@gvsu.edu. Please note that I cannot provide accommodations based upon disability until I have received a copy of the DSR issued memo. All discussions will remain confidential. For more information, see https://www.gvsu.edu/dsr/

Computing Success Center: he Computing Success Center, located in MAK A-1-101, is a great place to get help or simply hang out. The Computing Success Center offers free drop-in tutoring for a variety of College of Computing courses throughout the week during the academic year.